

Exam Date & Time: 27-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

SYSTEM ON CHIP DESIGN [ECE-4087]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1)
 - A) A processor die (1.4cm X 1.4cm) will be produced for five years. Over this period, defect densities are expected to drop linearly from 1.5 defects/cm² to 0.8 defects/cm². The cost of 6-inch wafer production will fall linearly from \$5,000 to \$3,500, and the cost of 8-inch wafer production will fall linearly from \$10,000 to \$6,500. Assume production of good devices is constant in each year. Which production process should be chosen? (5)
 - B) Design a finite state machine for a serial binary input which is divisible by 5. (3)
 - C) What is pipelining in a processor? What are different hazards which can be encountered during pipelining? (2)
- 2)
 - A) Design a Datapath for dividing two numbers input by the user, using adder/subtraction, comparator units. Identify the high-level state diagram for the machine. (5)
 - B) What are the different constituent units of SoC? (3)
 - C) Explain Snoopy and MESI cache protocols with a neat diagram. (2)
- 3)
 - A) (4)

A four-segment pipeline implements a function and has the following delays for each segment($b=0.2$):

Segment#	Maximum delay*
1	1.7ns
2	1.5ns
3	1.9ns
4	1.4ns

*Excludes clock overhead of 0.2ns

- What is the cycle time that maximizes performance without allocating multiple cycles to a segment?
- What is the total time to execute the function (through all stages)
- What is the cycle time that maximizes performance if each segment can be partitioned into sub systems?

- B) Use K-mean algorithm to cluster the following 7 examples into 2 clusters: $A_1 = (185, 72)$, $A_2 = (170, 56)$, $A_3 = (168, 60)$, $A_4 = (179, 68)$, $A_5 = (182, 72)$, $A_6 = (188, 77)$, $A_7 = (180, 71)$. (4)

- C) Explain the building blocks of NoC with a neat diagram. (2)

- 4) What is arbitration? Explain the difference between priority based and TDMA based arbitration protocols. (5)

- A) What is the significance of routing in NoC? Explain any one routing algorithms of NoC. (3)

- C) Differentiate between 2-D mesh and torus topologies of NoC. (2)

- 5) Find the minimum number of Test vectors to detect stuck at faults at all the given points (a, b, c, d, f, g, h) of the circuit shown in Fig.5a

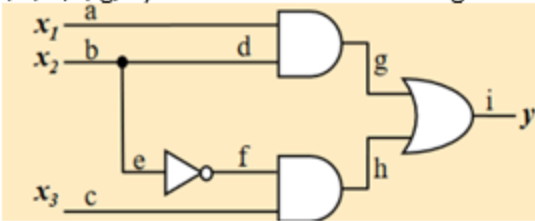
- A)  (5)

Fig.5a

- B) What is clock tree synthesis? Draw the structures of different types of clock trees. (3)

- C) Classify the IP's based on size and explain them. (2)

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